

RESEARCH

Open Access



A case study on the calculation of carbon emissions over the entire life cycle of commercial office buildings based on measured data

Meijiu Zhao¹ and Xiaochun Zhao^{1*}

*Correspondence:

Xiaochun Zhao

zhaomjiu1986@163.com

¹Hainan Vocational University of
Science and Technology,
Haikou 570000, China

Abstract

Against the backdrop of the “dual carbon” strategy, building carbon emissions have been incorporated into the construction review process. Currently, the calculation and accounting of carbon emissions throughout the entire building lifecycle have become a hot and challenging issue in the field of building carbon emissions. Commercial office buildings, due to their large scale, high energy consumption, and significant carbon emission base, have become a key area for energy conservation and carbon reduction in public buildings. According to the building lifecycle carbon emission assessment system, the lifecycle of commercial office buildings can be divided into four stages: production of building materials, construction, operation and maintenance, and dismantling and recycling. This study takes an existing commercial office building in Beijing, China, as a case study, and based on data from energy audit reports, calculates the carbon emissions of each lifecycle stage using national standards and relevant software, and discusses the factors affecting building carbon emissions. At the same time, a comparative analysis of the differences between Chinese and Western national standards is conducted. Ultimately, strategies to reduce building carbon emissions are proposed. This study is of reference value for the precise calculation of carbon emissions throughout the lifecycle of commercial office buildings.

Keywords Commercial office building, Building carbon emissions, Carbon emission calculation

Introduction

In September 2020, President Xi Jinping of the People’s Republic of China delivered an important speech at the 75th United Nations General Assembly, promising that “China will increase its national independent contribution and adopt more powerful policies and measures to strive to peak CO₂ emissions before 2030 and achieve carbon neutrality before 2060” [1]. The “dual carbon” strategic goal has clarified the future development

goals of my country's carbon emissions, and has posed a particular challenge to the construction industry.

The "2023 China Building and Urban Infrastructure Carbon Emissions Research Report" [2] jointly released by the China Building Energy Conservation Association and Chongqing University shows that in 2021, the total carbon emissions from the entire process of housing construction in China, including production, construction and demolition, construction, and operation, will be 4.07 billion tons of CO₂, accounting for 38.2% of the country's energy-related carbon emissions. Among them, the carbon emissions in the operation stage account for the largest proportion, accounting for 56.6% of the total carbon emissions. Therefore, exploring ways to reduce carbon emissions during the operation of buildings is an important task for my country to achieve its carbon peak and carbon neutrality goals.

There are three main methods for calculating carbon emissions in the building sector, namely: inventory method, empirical coefficient method and proportion method. Among them, the most widely used is the inventory method, which mainly uses carbon emission factors for measurement. At present, software simulation is still an important method for calculating carbon emissions. Some scholars have discussed the differences between simulated energy consumption and measured energy consumption through empirical research and proposed corresponding optimization strategies. Some scholars have also used a combination of Energy Plus and GIS to calculate building energy consumption and evaluated the impact of building morphology indicators on building energy consumption through regression analysis. Unfortunately, the existing research objects all focus on built-up residential areas, and the internal environment is heterogeneous, so the research results are difficult to apply to the carbon emission accounting of newly built large-scale affordable housing [3].

At present, research on residential building carbon emissions in China is mainly focused on the eastern and southern regions, where human activities are frequent and carbon emissions are relatively large. In contrast, there is relatively less attention given to the western and northern regions. However, the increase in carbon emissions and the rise in environmental costs have a more profound impact on less developed areas. As one of the most developed, economically largest, densely populated, and dynamic regions in China, the North China region is particularly prominent in terms of residential building carbon emissions and fossil fuel carbon emissions. There are differences in carbon emission characteristics between the North China region and the South China region, thus necessitating targeted research [4, 5].

As the largest economic center in northern China and the most densely populated area, Beijing attaches great importance to the application of emerging technologies and the concept of low-carbon construction and living, reflecting the high level of contemporary urban planning and design in our country. This article takes a commercial office building that has been completed and put into use in Beijing as an example to conduct research on the calculation of carbon emissions throughout the building's life cycle, aiming to respond to the national policy guidance on green and low-carbon development. Against the backdrop of increasingly severe global climate change and environmental issues, the Chinese government has put forward the goals of "carbon peak" and "carbon neutrality", that is, striving to achieve peak carbon emissions by 2030 and carbon neutrality by 2060. To achieve this grand goal, the Chinese government has clearly stated in

the “14th Five-Year Plan” and the long-term goals for 2035 that it is necessary to promote green and low-carbon development and accelerate the construction of an economic system that is green and low-carbon and circular. In such a policy context, Beijing, as the capital of China and an international metropolis, it is particularly important to integrate the concept of low-carbon into urban construction and management. As an important part of urban energy consumption and carbon emissions, the research on the calculation of carbon emissions throughout the life cycle of commercial office buildings is of great significance for promoting the low-carbon transformation of the construction industry. By constructing a carbon emission analysis model for the entire life cycle of commercial office buildings, the inherent logic and source of carbon emissions over the entire life cycle of commercial office buildings can be accurately understood, and optimization suggestions can be put forward. This study has reference value for the accurate calculation of carbon emissions over the entire life cycle of commercial office buildings.

Methodology

Project overview

The case study object is a commercial office building located in Xicheng District, Beijing. The building was built in 2012. The building has a construction area of 112,985 m², 25 floors above ground, 4 floors underground, and a height of 98.8 m. It is a frame structure. The heat source of the building is a gas boiler. The cold source of the building is an electric chiller. The main energy systems of the building include air conditioning system, lighting system, elevator system, etc. The specific parameters of the electric chiller are shown in Table 1. This table lists the parameters of two different models of electric water chillers (electric refrigeration units). These parameters include model name, number of units, type of refrigerant used, amount of refrigerant charged, the Global Warming Potential (GWP) value of the refrigerant, and the service life of the equipment.

Energy consumption data source

After the building was basically fully occupied in 2015, the property company recorded the various energy consumption of the building in detail every year. In 2018, the building was audited as an energy-saving demonstration project. In the following ten years, the data of the equipment operation ledger changed steadily with small fluctuations. Therefore, the data in the 2018 energy audit report was used as the basis for the calculation of the building's carbon emissions. According to statistics, the annual electricity consumption of the building in 2018 was 991.86kWh/a, the annual water consumption was 95,555 tons, and the annual natural gas consumption was 693,829m³/a. When conducting an energy audit on the building, a sub-item electricity metering system was installed in the distribution room of the building. The sub-item electricity consumption for the whole year is shown in Fig. 1, among which office, lighting, and air conditioning have the highest energy consumption, accounting for 75.05% of the total electricity consumption.

Table 1 Electric chiller parameters

Model	Number of units	Refrigerant	Filling amount	Global Warming Potential (GWP)	Service life
CHVG670-600-03	4	R123	499 kg	79	20 year
RHBT255	1	R134a	238 kg	1300	20 year

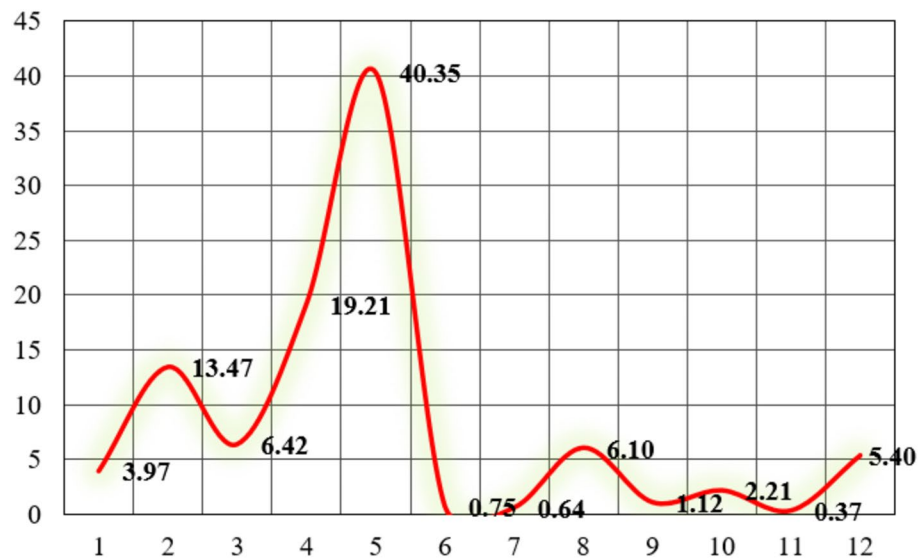


Fig. 1 Breakdown of building energy consumption throughout the year (total power consumption 991.86kWh/a)

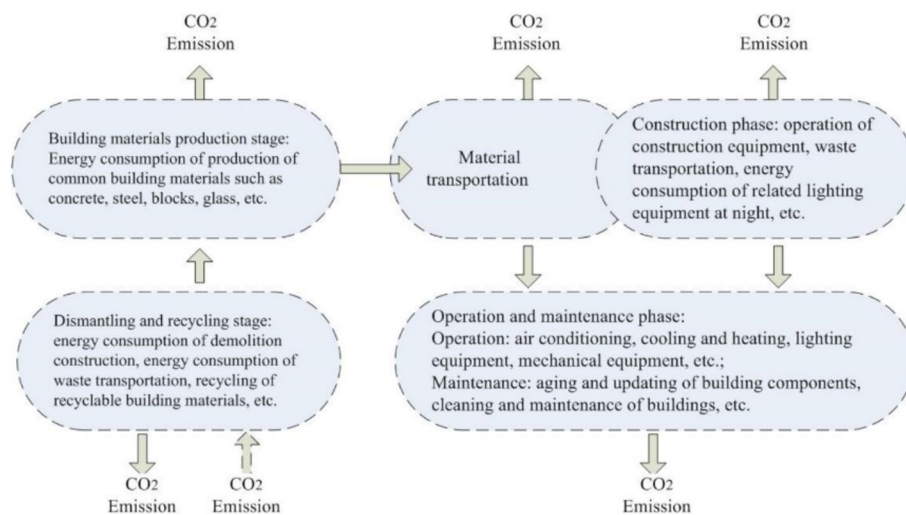


Fig. 2 Calculation scope of building carbon emissions

The unit of the vertical axis is %; the horizontal axis represents each sub-item. 1-Others; 2-Air conditioning; 3-Information equipment; 4-Lighting; 5-Office; 6-Special function equipment; 7-Elevator; 8-Heating power consumption; 9-Exhaust system; 10-Transformer loss; 11-Water supply and drainage; 12-Split air conditioning.

Calculation boundary of building carbon emissions

The Ministry of Housing and Urban-Rural Development of the People's Republic of China has issued and implemented the "Building Carbon Emission Calculation Standard" (GB/T51366-2019) [5]. The "Building Carbon Emission Calculation Standard" divides the entire life cycle of a building into three stages: building material production and transportation, building operation, and building construction and demolition, as shown in Fig. 2. The calculation of building carbon emissions should include all types of greenhouse gases listed in the "IPCC National Greenhouse Gas Inventory Guidelines".

The time range should be consistent with the design documents. When the design documents are not available, the calculation is based on 50 years. Carbon emissions should take into account the carbon sink reduction of building green spaces [6].

The types of energy consumption that are taken into account in the calculation encompass a wide range of sources. These include coal, oil, and natural gas, which are traditional fossil fuels commonly utilized in various forms of power generation systems. Additionally, renewable energy sources such as biomass energy, wind energy, and solar energy are also considered. These are increasingly being integrated into central heating systems and central cooling systems, which are essential for maintaining comfortable environments in residential and commercial buildings. Furthermore, the calculation also accounts for the energy consumption in the form of electricity, gasoline, and diesel that is used during the construction phase of projects. This phase includes the energy required for the operation of construction machinery, transportation of materials, and other activities related to the building process.

Calculation method of building carbon emissions

Calculation formula

According to the Standard, the total carbon emissions during the life cycle of a building are calculated as follows:

$$C = C_M + (C_{JZ} + C_{CC}) + C_{JC} \quad (1)$$

where C is the total carbon emissions per unit area during the entire life cycle of the building, in kgCO_2/m^2 ; C_M , C_{JZ} , C_{CC} , C_{JC} are the carbon emissions per unit area generated during the building operation, construction, demolition, building materials production and transportation stages, in kgCO_2/m^2 ; " $\text{kgCO}_2\text{e}/\text{m}^2$ " is a unit that represents the greenhouse gas emissions per unit area, where " kgCO_2e " stands for kilogram of carbon dioxide equivalent ("e" is the abbreviation for "equivalent"), and " m^2 " represents square meters.

The carbon emissions of each stage of the building are calculated using the emission factor method [7]. Among them, the carbon emissions in the operation stage include the carbon emissions generated by the energy consumption of systems such as HVAC, domestic hot water, and lighting (minus the capacity of renewable energy systems) and the calculation of the carbon reduction of building carbon sinks. The key to carbon emissions in the construction and demolition stages is to determine the consumption of various types of energy in the construction stage. The carbon emissions in the building materials production and transportation stage include the carbon emissions generated in the upstream raw materials of building materials, the mining of energy, the entire process of building materials production, and the transportation of building materials out of the factory.

In order to facilitate the understanding of the carbon emissions of the building throughout its life cycle, the accounting stage, accounting boundary and accounting indicators of this case study adopt the carbon emission calculation cycle (the entire life cycle of the building), calculation boundary, calculation rules, accounting rules and carbon emission factors stipulated in the "Building Carbon Emission Calculation Standard".

(1) Calculation of carbon emissions in the building materials production and transportation stage.

The carbon emissions in the building materials production and transportation stage should be the sum of the carbon emissions in the building materials production stage and the carbon emissions in the building materials transportation stage, calculated according to formula (2). Among them, the carbon emissions in the building materials production stage are calculated using formula (3), and the carbon emissions in the building materials transportation stage are calculated using formula (3).

$$C_{JC} = \frac{C_{sc} + C_{ys}}{A} \quad (2)$$

$$C_{sc} = \sum_{i=1}^n M_i F_i \quad (3)$$

$$C_{ys} = \sum_{i=1}^n M_i \times D_i \times T_i \quad (4)$$

where C_{JC} : Carbon emission per unit area in the production and transportation of building materials, $\text{kgCO}_2\text{e}/\text{m}^2$; C_{sc} : Carbon emission in the production of building materials, kgCO_2e ; C_{ys} : Carbon emission in the transportation of building materials, kgCO_2e ; A : Building area, m^2 ; M_i : Consumption of the i -th main building material, t ; F_i : Carbon emission factor of the i -th main building material, $\text{kgCO}_2\text{e}/\text{unit quantity of building materials}$; D_i : Average transportation distance of the i -th building material, km ; T_i : Carbon emission factor per kilometer of transportation distance per unit weight under the transportation mode of the i -th building material, $\text{kgCO}_2\text{e}/(\text{t}\cdot\text{km})$.

(2) Calculation of carbon emissions during the construction phase.

The carbon emissions during the construction phase of a building should be calculated based on the various fuel and power usage during the construction phase and the corresponding energy carbon emission factors. Calculate according to formula (5):

$$C_{JZ} = \frac{\sum_{i=1}^n E_{jz,i} EF_i}{A} \quad (5)$$

In the formula, C_{JZ} : total carbon emissions during the construction process, kgCO_2/m^2 ; $E_{jz,i}$: total consumption of the i -th fuel power during the construction process, kWh or kg ; EF_i : carbon emission factor of the i -th fuel power, kgCO_2/kWh or kgCO_2/kg ; A : building area, m^2 .

The total energy consumption during the construction phase and the total energy consumption of each sub-item and sub-project are calculated in accordance with the relevant provisions of the "Building Carbon Emission Calculation Standard".

(3) Calculation of carbon emissions during the operation phase.

The process of calculating carbon emissions that occur during the operational phase of a building should encompass a thorough evaluation of the total carbon emissions stemming from various systems, including but not limited to the heating of the building, the cooling systems, the provision of domestic hot water, the lighting systems, and the integration of renewable energy systems throughout the operational lifecycle of the building [8].

It is imperative that the lifespan of the building aligns with the specifications outlined in the design documentation, and for this purpose, it is to be calculated as spanning

a period of 50 years. The scope of the building's carbon emission calculations should encompass all carbon emissions resulting from the energy consumption that falls within the boundaries of the building project planning license. Additionally, it should factor in the carbon reduction benefits derived from renewable energy sources and carbon sink systems that are part of the building's infrastructure.

To ascertain the carbon emissions of each system within the building, it is necessary to base the determination on the various types of energy consumption specific to each system, along with the corresponding carbon emission factors for each type of energy utilized. The carbon emissions per unit area of the building throughout its operational phase should be meticulously calculated using formula (6) and formula (7):

$$C_M = \frac{[\sum_{i=1}^n (E_{jz,i} EF_i) - C_p]y}{A} \quad (6)$$

$$E_i = \sum_{j=1}^n (E_{i,j} - ER_{i,j}) \quad (7)$$

In the formula, C_M : carbon emissions per unit building area during the building operation phase, kgCO_2/m^2 ; E_i : annual consumption of the i -th type of energy in the building, unit/a; EF_i : carbon emission factor of the i -th type of energy, kgCO_2/kWh or kgCO_2/kg ; $E_{i,j}$: consumption of the i -th type of energy by the j -th type of system, unit/a; $ER_{i,j}$: consumption of the i -th type of energy provided by the renewable energy system by the j -th type of system, unit/a; i : type of terminal energy consumed by the building, including electricity, gas, oil, heat, etc.; j : type of building energy system, including air conditioning, lighting, etc.; C_p : annual carbon reduction of the building green space carbon sink system, kgCO_2/m^2 ; y : design service life, a; A : building area, m^2 .

Special note: renewable energy systems should include solar domestic hot water, photovoltaic systems, ground source heat pump systems and wind power generation systems. For details, please refer to the "Building Carbon Emission Calculation Standard".

(4) Calculation of carbon emissions during the demolition phase.

The carbon emissions during the building demolition phase should be calculated according to the various fuel and power consumption during the demolition phase and the corresponding energy carbon emission factors, using formula (8):

$$C_{CC} = \frac{\sum_{i=1}^n E_{CCi} \times EF_i}{A} \quad (8)$$

In the formula, C_{CC} : total carbon emissions during building demolition, kgCO_2/m^2 ; $E_{cc,i}$: total consumption of the i -th fuel power during building demolition, kWh or kg ; EF_i : carbon emission factor of the i -th fuel power, kgCO_2/kWh or kgCO_2/kg ; A : building area, m^2 .

The energy consumption during the manual and mechanical demolition stages of buildings is calculated according to formula (9) and formula (10):

$$E_{CC} = \sum_{i=1}^n Q_{cc,i} f_{cc,i} \quad (9)$$

$$f_{cc,i} = \sum_{j=1}^m T_{B-i,j} R_j + E_{jj,i} \quad (10)$$

where E_{cc} : energy consumption in the building demolition process, kWh or kg; $Q_{cc,i}$: the engineering quantity of the i th demolition project; $f_{cc,i}$: energy consumption coefficient per measurement unit of the i th demolition project, kWh/engineering measurement unit or kg/engineering measurement unit; $T_{B-i,j}$: energy consumption per unit of engineering quantity of the j th type of construction machinery in the i th demolition project; R_j : energy consumption per unit of construction machinery in the j th type of construction machinery in the i th demolition project; i : project number in the demolition project; j : construction machinery number.

Carbon emission factor

The carbon emission factor refers to the coefficient corresponding to the energy and material consumption and carbon dioxide emissions. The carbon emission factors of the main energy sources (fossil fuels and other energy sources), building materials and building material transportation commonly used in the construction field are given in the appendix of the standard “Building Carbon Emission Calculation Standard” [9].

Carbon emissions accounting

The calculation and accounting of carbon emissions during the life cycle of a building is a hot and difficult issue in the field of building energy conservation at this stage. The calculation is usually carried out in the design stage to predict the carbon emissions of a building during its life cycle based on the design plan or design drawings. The purpose is to compare and select design plans. The accounting of building carbon emissions is used for completed buildings and is based on the building energy consumption metering data. The results are basically real carbon emission data.

In the specific calculation process of this study, the carbon emission calculation and analysis software PKPM was used. The software uses the carbon emission calculation kernel of the tool (IBE) approved by the “Building Carbon Emission Calculation Standard” [10]. A three-dimensional model of the project was established according to the design drawings and imported into the PKPM software for analysis. In the building operation stage, this paper calculated the carbon emissions of system energy consumption, the carbon emissions simulated by the software, and the carbon emissions based on energy ledger data. Due to the lack of relevant data on construction energy consumption, the construction process energy consumption estimation method recommended by the “Building Carbon Emission Calculation Standard” and the software calculation method were used in the construction and demolition stages of the building. The carbon emissions of this stage were calculated using these two methods. In the production and transportation stage of building materials, the carbon emissions calculation method specified in the “Building Carbon Emission Calculation Standard” was used.

(1) Operational stage.

According to the “Building Carbon Emission Calculation Standard”, this study calculates the energy consumption of the building’s HVAC, domestic hot water, lighting and elevator systems, and then obtains the carbon emissions of each system. The results are detailed in Table 2.

Table 2 Calculation results of carbon emissions of various systems during the building operation phase

Operating system	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² ·a)
Air conditioning system	81519.77	727.96	1630395.30	14.56
Heating and hot water system	117967.67	1053.44	2359353.55	21.07
Lighting system	91745.23	819.27	1834904.52	16.39
Elevator system	3581.93	31.99	71638.64	0.64
Total	294814.60	2632.65	5896292.01	52.66

Table 3 Calculation results of carbon emissions of each system during the building operation phase (calculated by software)

Operating system	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² ·a)
Air conditioning system	54214.51	484.13	1084290.24	9.68
Heating and hot water system	53674.59	479.30	1073491.76	9.58
Lighting system	60278.16	538.27	1205563.26	10.77
Elevator system	2353.40	21.01	47067.90	0.42
Total	20145.02	179.89	402900.30	3.60

Table 4 Carbon emissions from various energy consumption during the building operation phase (based on energy consumption data from energy audits)

Operating system	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² ·a)
Gas	82524.86	736.93	1650497.20	14.74
Electricity	477590.87	4264.81	9551817.44	85.29
Tap water	4781.48	42.70	95629.53	0.86
Refrigerant	1284.48	11.48	25689.62	0.23
Total	566181.69	5055.92	11323633.79	101.12

In order to further explore the influencing factors of carbon emissions in the building operation stage, this study uses the carbon emission calculation and analysis software PKPM for simulation calculation. The calculation results are shown in Table 3.

Tables 2 and 3, which are presented in this document, encompass detailed design calculations. The primary source of carbon emissions for this particular building stems predominantly from the energy inputs required for its operation. These inputs include natural gas, electricity, and tap water. Additionally, there is a significant contribution to the carbon footprint from non-CO₂ greenhouse gases that are emitted by the refrigerants used in the building's chillers. To ascertain the precise carbon emissions associated with the various forms of energy consumption throughout the operational phase of the building, a thorough verification process has been conducted. This process is based on the energy consumption ledger data obtained from the energy audit. The outcomes of these calculations are meticulously recorded and presented in Table 4, which provides a comprehensive overview of the carbon emissions for each type of energy consumption.

(2) Construction and demolition phase.

This study proposes a method for estimating energy consumption during the construction process of buildings, aimed at calculating the carbon emissions during the construction and demolition phases. First, the main quantities of work are accounted for, and then, based on the national standard “Quota Standard for Consumption of House Building and Decoration Engineering” (TY01-31-2015) [11], the mechanical unit consumption and energy consumption per unit of work are calculated, thereby estimating the carbon emissions of the building during the construction and demolition phases. The calculation results are detailed in Table 5.

The usage and energy consumption data of construction machinery, tools and equipment are usually difficult to obtain, and the calculation process is relatively complicated. The software PKPM proposes a simple algorithm. According to the empirical formula (11), the carbon emissions per unit area of the building are estimated.

$$Y = X + 1.99 \quad (11)$$

where Y: Carbon emissions per unit area; X: Number of above-ground floors of the building.

This particular method is exclusively designed for engineering projects that lack any form of data pertaining to energy consumption specifically related to the construction phase of the building. Despite the absence of such data, it remains essential to ascertain the carbon emissions that occur during this critical phase of construction. To achieve this, the method calculates the carbon emissions per unit area of the building during the construction phase, which has been determined to be 30.79kgCO₂e/m².

When it comes to the building demolition phase, the software PKPM advises the adoption of the building volume estimation method. This approach is based on empirical formula (12), which allows for the estimation of carbon emissions per unit area of the building during the demolition phase. This estimation is crucial for understanding the environmental impact of the demolition process and for taking steps to mitigate it.

$$Y = 0.06X + 2.01 \quad (12)$$

Through calculation, it can be obtained that the carbon emissions per unit area of the building during the demolition phase is 3.93kgCO₂e/m². The calculation results are shown in Table 6.

(3) Building materials production and transportation stage.

There are many types of energy, materials, and machinery involved in the production process of building materials. Different companies have different classification standards and measurement units for the same building materials. For example, waterproofing membranes are measured in two ways: m² and kg. The measurement unit of concrete is m³, but the emission factor unit for transportation is kgCO₂e/(t•km). Moreover, the

Table 5 Carbon emissions during the construction and demolition phases of buildings

Construction phase	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² •a)
Construction phase	3749.39	33.48	74987.88	0.67
Demolition phase	317.81	2.84	6356.24	0.06
Construction and demolition phase	4067.21	36.32	81344.12	0.73

Table 6 Calculation results of carbon emissions during the construction and demolition phases of buildings (calculated by software)

Construction phase	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² ·a)
Construction phase	3447.87	30.79	68957.50	0.62
Demolition phase	439.76	3.93	8795.16	0.08
Construction and demolition phase	3887.63	34.72	77752.62	0.69

Table 7 Calculation results of carbon emissions in the production and transportation stages of building materials

Construction phase	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² ·a)
Production phase	50050.96	446.95	1001019.14	8.94
Transportation phase	5625.49	50.24	112509.76	1.00
Production and transportation phase	55676.45	497.18	1113528.90	9.94

Table 8 Calculation results of carbon emissions in the production and transportation stages of building materials (calculated by software)

Construction phase	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²	Carbon emission intensity kgCO ₂ e/a	Carbon emission intensity kgCO ₂ e/(m ² ·a)
Production phase	53479.02	477.55	1069580.38	9.55
Transportation phase	6315.93	56.40	126318.50	1.13
Production and transportation phase	59794.94	533.95	1195898.88	10.68

database of carbon emission factors for building materials is still incomplete. For example, concrete is divided into 19 grades, but the “Building Carbon Emission Calculation Standard” only gives the carbon emission factors of C30 and C50 concrete [12].

When conducting the calculations in accordance with the “Building Carbon Emission Calculation Standard,” it is important to note that the carbon emission factors for waterproofing membranes and tempered glass are sourced from the “China Product Life Cycle Greenhouse Gas Emission Coefficient Set (2022).” All other factors are derived from the appendix of the “Building Carbon Emission Calculation Standard.” The outcomes of these calculations are then presented in two separate tables for comparison. Table 7 displays the results obtained by adhering strictly to the “Building Carbon Emission Calculation Standard,” while Table 8 showcases the outcomes derived from utilizing the software PKPM. This dual presentation allows for a comprehensive analysis and verification of the data, ensuring accuracy and reliability in the assessment of carbon emissions for building projects.

Results & discussion

Carbon emissions during the entire life cycle of a building and the proportion of each stage

The carbon emissions and proportions of the building at each stage of its life cycle obtained based on the above calculation method are shown in Table 9. The software

Table 9 Comparison of total carbon emissions at each stage of the building's life cycle

Calculation method	Total carbon emissions tCO ₂ e	Carbon emission intensity kgCO ₂ e/m ²
Total carbon emissions tCO ₂ e	354558.25	257862.65
Carbon emission intensity during operation kgCO ₂ e/(m ₂ ·a)	52.66	34.68
Carbon emission intensity during construction and demolition kgCO ₂ e/(m ₂ ·a)	0.73	0.69
Carbon emission intensity during building materials production and transportation kgCO ₂ e/(m ₂ ·a)	9.94	10.68
Annual average carbon emission intensity kgCO ₂ e/(m ₂ ·a)	63.33	46.06

Table 10 Summary of calculation cases of carbon emissions of commercial office buildings throughout their life cycle

Authors	Years	Area	Annual average carbon emission intensity kgCO ₂ e/(m ² ·a)	Carbon emissions during use	Carbon emissions from building materials production and transportation	Carbon emissions during construction and demolition
Lin Yuedong et al. [14]	2021	Xiamen	65.55	82.62%	15.85%	1.53%
Li Yuanzhao et al. [15]	2021	Tianjin	48.30~66.10	89.30%	10.40%	0.30%
Bai Luheng [16]	2019	Tianjin	40~90	89.30%	10.35%	0.35%
Chen Yi [13]	2019	Tianjin	48.80	75.59%	23.72%	0.69%
Dong Kaihong [17]	2018	Foshan	108.65	89.34%	8.93%	1.73%

calculation results are basically the same as the building energy consumption calculated according to the Building Carbon Emission Calculation Standard, with a deviation of less than 10%. The carbon emission factors and database in the software are constantly updated. For example, the carbon emission factor of electricity is decreasing year by year, and the carbon emission calculation results obtained using the latest data are closer to the actual level [13].

As depicted in Table 9, throughout the entire life cycle of a building, it is evident that the carbon emissions associated with the operational phase are by far the most significant, constituting over 75% of the total. Following this, the production and transportation of building materials contribute the second highest amount of carbon emissions, which accounts for more than 15%. In contrast, the carbon emissions generated during the construction and demolition phases of the building are relatively minor and can, for the most part, be disregarded. The findings presented in this paper are in line with the proportions of carbon emissions calculated for each stage of office buildings by other researchers in the field. As shown in Table 10. The proportion of carbon emissions in each stage of the whole life cycle of commercial office buildings in different regions is relatively small, but the annual average carbon emission intensity is quite different. The reason may be that the building structure, type of building materials, equipment operation and building energy consumption are different.

Boundary demarcation

By comparing Tables 2 and 3, and Table 4, it can be observed that during the building operation phase, the carbon emissions calculated directly based on energy consumption are significantly higher than the results obtained according to the “Building Carbon

Table 11 Scope of application and main contents of the standard

Scope of application	Main content
CM1	Central heating/cooling and hot water, fresh air system. Fixed lamps and electrical appliances, plug-in mobile heating and cooling equipment and lamps, building elevators and auxiliary equipment, etc.
CM2	On the basis of CM1, add auxiliary plug-in lamps for special use by computing stations or users, and other household or office appliances (such as printers, televisions). Special equipment such as kitchen appliances, refrigerators and freezers, data center-specific equipment, and other special function devices.
CM3	All elements within the system boundary of CM2 and other processes and activities (including upstream and downstream processes), including greenhouse gas emissions and removals related to the use phase of the building and other systems in the building site caused by these processes and activities. It also includes major maintenance, such as cleaning, repair, replacement and renovation, water use, waste treatment and disposal, and refrigerant emissions from building air conditioning systems.

Emission Calculation Standard” and corresponding software. The reason for this discrepancy may lie in the fact that the “Building Carbon Emission Calculation Standard” does not take into account the carbon emissions from office electrical appliances and equipment. As shown in Fig. 1, office electricity consumption accounts for 40.35% of the total electricity consumption, which is roughly equivalent to the electricity consumption of the computing equipment system, with the corresponding carbon emission intensity being 31.25kgCO₂e/(m²•a).

The international standard known as “Sustainability of buildings and civil engineering projects - Carbon emissions from existing buildings during the use phase” (Part 1: Calculation, reporting and communication) (ISO16745-1-2017) [18] sets forth a universally recognized approach for quantifying and reporting on the greenhouse gas emissions originating from existing structures throughout their operational phase. This encompasses the measurement, reporting, and verification of carbon emissions during the use of buildings. The standard’s applicability includes an examination of both direct and indirect carbon emissions that are associated with buildings. These emissions are categorized into three distinct groups. CM1 pertains to the computation of carbon emissions that are directly linked to the energy consumption of the building itself. CM2 involves the assessment of carbon emissions that are a consequence of the building’s own energy use, as well as the energy consumption that is associated with the activities of its occupants. CM3 expands upon CM1 and CM2 by also taking into account other greenhouse gases that are produced during the operation of the building. This includes, but is not limited to, direct and indirect carbon emissions resulting from activities such as building cleaning, maintenance, and renovation, along with emissions from refrigerants. The detailed information regarding these categories is further elaborated in Table 11.

The “Building Carbon Emission Calculation Standard” holds that the use of household appliances, office appliances, cooking, and other usage patterns have a significant impact on them. Therefore, it is not recommended to include the carbon emissions caused by them in the calculation of building carbon emissions. The “Building Carbon Emission Calculation Standard” only stipulates the calculation of carbon emissions corresponding to CM1 category.

The project calculation utilized energy ledger data, including all computers, water dispensers, printers, and other office equipment within the building, which fall under the CM2 category. The sub-metering system installed during the energy audit process indicates that this part of the electricity consumption is relatively high. Therefore, this has

a significant impact on the calculation of building carbon emissions belonging to the CM2 category. Based on over ten years of sub-metering system trial experience, the electricity consumption of workstation equipment is also difficult to separate and deduct in actual measurement data, thus making the measurement and statistics of building carbon emissions belonging to the CM2 category easier to implement.

The project calculation also includes refrigerant emissions. This falls under the CM3 category but does not include carbon emissions from other maintenance and renovation activities. The proportion of these results is not high, but they are closely related to actual use and maintenance. According to the property management company, the refrigeration unit does not involve refrigerant emissions. Therefore, if based on a 20-year service life, with refrigerant being charged and discharged every 5 years, carbon emissions would increase fourfold. Thus, the impact of equipment manufacturing and maintenance on carbon emissions cannot be ignored.

Stage division

During the operational phase of a building, calculating the energy consumption and other data of existing buildings through actual records can avoid the complexity and uncertainty of the estimation process. The results of these calculations are more realistic and facilitate comparative analysis. The carbon emissions during the operational phase of a building account for over 75% of the total carbon emissions throughout the building's lifecycle. Therefore, it is recommended that when calculating the carbon emissions of buildings in our country, the focus should primarily be on assessing the carbon emissions during the operational period of the buildings.

During the construction and demolition phases of buildings, the energy consumption calculation method recommended by the "Building Carbon Emission Calculation Standards" is cumbersome, requiring the statistics of energy consumption data for all construction tools, which is inconvenient and prone to omissions. Drawing on the experience from the operational phase, the overall statistical data of temporary energy consumption (electricity, water, and gas) at the construction site can be utilized to calculate the carbon emissions during the construction phase of buildings, thereby reducing the complexity of the current calculation method.

In addition, due to the long time span, the demolition phase can generally only be estimated. For existing buildings, when there is a lack of energy consumption data, it is recommended to use the above empirical formulas (11) and (12) for estimation. In addition, the carbon emissions during this phase account for a small proportion. From a management perspective, it is recommended that the energy consumption and carbon emissions of this part can be included in the scope of building construction management.

During the production and transportation of building materials, it is necessary to count the amount of different building materials and the distance of transportation vehicles, and the database of relevant carbon emission factors needs to be continuously improved. This method also has the problems of cumbersome and complex calculations and incomplete database data. From the perspective of national statistics, it also belongs to different fields such as industry (cement, steel and other industries) and transportation. It is recommended that the carbon emissions of production and transportation should be considered separately.

Carbon reduction strategy

Through case analysis, it is found that the main factors affecting building carbon emissions are energy consumption and building technology. Carbon emissions can be reduced from the following three aspects.

(1) Reduce energy consumption. The commercial office building sector is responsible for a significant portion of carbon emissions during the operational phase, which accounts for approximately 80–90% of the total lifecycle emissions of a building. The average commercial office building in China consumes about 100–150 kWh/m² annually. By implementing energy-saving measures, such as upgrading to high-efficiency HVAC systems that can reduce energy consumption by up to 30%, and optimizing lighting and elevator systems to save an additional 20%, the total energy consumption can be cut by approximately 25–35%. This translates to a reduction in carbon emissions of about 25–35 kg CO₂/m² per year, based on the average energy mix in China. The adoption of these measures aligns with the national building energy conservation standards, which aim to improve energy efficiency and significantly contribute to the reduction of carbon emissions.

(2) Improve building technology. Building energy conservation has been implemented in the management of the entire process from design to construction. Therefore, new dimensions need to be added in the selection of building materials and structures, including optimizing building form, improving assembly, adopting high-efficiency energy-saving building materials, and improving the recycling rate of building materials. However, this part of the content still needs to be continuously improved by the database of building materials and production.

(3) Actively utilize renewable energy. Introducing and applying renewable energy into buildings is an important direction for the future development of the construction industry. Vigorously promote the development and utilization of renewable energy power generation. Actively expand the scale of non-electric utilization of renewable energy. As the proportion of renewable energy in my country's power system continues to increase, its carbon emission factor continues to decrease. Therefore, building electrification is more conducive to emission reduction than directly burning primary energy. In addition, making full use of solar energy, wind energy, geothermal energy and other resources as much as possible and reducing the use of non-renewable energy will help reduce carbon emissions.

Conclusion

- (1) During the entire life cycle of a building, the carbon emissions during the operation phase account for more than 75%, becoming the core focus of energy conservation and carbon reduction efforts in buildings. The method of calculating carbon emissions based on energy consumption data during the operation phase is relatively simple and reliable, in line with international standards, and is conducive to horizontal comparisons.
- (2) The calculation of carbon emissions during the construction, demolition, production, and transportation of building materials involves a vast amount of content, with complex statistical projects, and the basic database urgently needs improvement. In the case of incomplete existing project data, it is recommended to use empirical formulas for direct calculation.

- (3) By calculating the carbon emissions of commercial office buildings, technologies and measures can be specifically selected to reduce carbon emissions, which will help achieve the dual-carbon goals in the field of construction.
- (4) Climate significantly affects building energy use and carbon emissions. Cold areas need more heating, while warm areas require more cooling. Choosing suitable materials and energy-saving methods is crucial. Building practices impact energy use and emissions, and employing green materials, energy-efficient designs, and advanced technologies can lower emissions. Enhancing building efficiency and minimizing waste are equally important. The energy grid's cleanliness and efficiency dictate building emissions, and using renewable energy sources like wind and solar can reduce them. Improving grid efficiency and cutting transmission losses are also vital.

Author contributions

Data collection, analysis and interpretation of results, draft manuscript preparation: Meijiu Zhao; data collection, critical review of the manuscript: Xiaochun Zhao. All authors reviewed the manuscript.

Funding

This research was funded by 2023 school level research project of Hainan Vocational University of Science and Technology: "Design and performance study of low carbon concrete partition panel and block based on plant fiber", grant number HKKY2023-ZD-04.

Data availability

All data generated or analysed during this study are included in this article.

Declarations

Competing interests

The authors declare that they have no competing interests.

Received: 20 August 2024 / Accepted: 26 April 2025

Published online: 13 May 2025

References

1. Pan Jiahua L, Meng Z (2021) Kun. China's plan to achieve carbon peak and carbon neutrality. *China Social Sciences Today*, (006)
2. China Building Energy Conservation Association (2021) China Building Energy Consumption and Carbon Emission Research Report: Provincial Building Carbon Peak Situation Assessment. 2021
3. United Nations Framework Convention on Climate Change (UNFCCC). The Paris Agreement. In: Proceedings of UN Climate Change Conference (2021) Oct31–Nov 13; Glasgow, Scotland. New York City: UNFCCC; 2021
4. China Academy of Building Research (2021) General code for Building energy conservation and renewable energy utilization (GB 55015–2021). China Construction Industry, Beijing
5. China Academy of Building Research (2019) Building carbon emission calculation standard (GB 51366–2019). China Construction Industry, Beijing
6. Guangdong Provincial Department of Housing and Urban-Rural Development. Guidelines for Calculation of Building Carbon Emissions (Trial) (2021) 12
7. United Nations Environment Programme (UNEP) (2021 Oct) 2021 Global status report for buildings and construction: towards a zero-emission, efficient and resilient buildings and construction sector. Report. UNEP, Kenya
8. Li XD, Zhu C (2020) A review of research on carbon emission accounting and influencing factors of buildings in my country. *J Saf Environ* 20(01):317–327. <https://doi.org/10.13637/j.issn.1009-6094.2019.0306>
9. Han YH, Yu Z, Chen T (2023) Carbon emission calculation and analysis of prefabricated buildings with different assembly rates based on carbon emission coefficient method. *Building Struct* 53(S1):1337–1342. <https://doi.org/10.19701/j.jzjg.23S1300>
10. Zheng LH, Feng CS (2014) Carbon emission accounting and energy saving and emission reduction benefit analysis of green Buildings throughout their life cycle - a case study of an office Building in Tianjin. *Dynamic (Ecological City Green Building)*, (03):60–62
11. Ministry of Housing and Urban-Rural Development (2015) Consumption quota for housing construction and decoration engineering (TY01-31-2015). China Planning, Beijing
12. Bai LH (2019) Research on prediction model of carbon emission of public buildings throughout their life cycle. Tianjin University
13. Chen Y, Song C, Xie PC, Dai XR (2019) Carbon emission analysis of green office buildings. *Constr Sci Technol*, (Z1):112–115
14. Lin YD, Shan PP, Huang W, Lin DX, Huang HB, Zhang CZ, Zhuo ZY (2021) Research on evaluation method of carbon emission of buildings in Fujian Province based on CIE-AHP. *Fujian Constr Sci Technol*, (02):77–80
15. Li YZ, Wu YT, Yu J, Bai LH, Wang LX (2021) SVR-based prediction model for carbon emissions of high-rise office buildings throughout their life cycle: A case study of Tianjin. *Building Energy Conserv (Chinese English)* 49(09):25–30

16. Bai LH (2019) Research on prediction model for carbon emissions of public buildings throughout their life cycle. Tianjin University
17. Dong KH (2018) Research on calculation of carbon emissions of large complex buildings throughout their life cycle in the South. South China University of Technology
18. ISO/TC 59 (2017) /SC 17 Sustainability in buildings and civil engineering works. ISO 16745-1:2017 sustainability in Buildings and civil engineering works carbon metric of an existing Building during use stage part 1: calculation, reporting and communication. BSI publications

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.